1895–96

20th-century botanists classified plant formations, such as vegetation on crumbling sand dunes (shown here), by their ecological characteristics.

THE SCIENCE OF PLANT ECOLOGY reached a milestone when Dutch botanist **Eugenius Warming** (1841–1924) and German botanist **Andreas Schimper** (1856–1901) published their books on the subject at the end of the century. Together, they showed how **vegetation could be classified** into different formations based on climate and soil conditions.

In Britain, physicists **Lord Rayleigh** (1842–1919) and **William Ramsay** (1852–1916) discovered the gas **argon**. Rayleigh realized that air must contain an unknown chemical component, since the density of atmospheric nitrogen was different from that of pure nitrogen made in a laboratory. He found that atmospheric nitrogen contained traces of argon and other unreactive elements later dubbed **noble gases**.

> “ I HAVE **SEEN MY DEATH.** ”
>
> **Anna Röntgen**, wife of Wilhelm, on seeing her hand X-ray, 1895

In November, German physicist **Wilhelm Röntgen** (1845–1923) discovered that electrically charged vacuum tubes emitted rays that made a fluorescent screen glow; he called them **X-rays**. He found that they went through human skin and exposed photographic plates. This led to the development of **medical radiography**. By 1896, scientists knew that X-rays could ionize (charge up) air. Some physicians even tried firing X-rays at tumors—**radiotherapy**—to try to cure cancer.

Inspired by Röntgen, French physicist **Henri Becquerel** (1852–1908) studied whether **phosphorescent substances**, such as uranium salts, produced X-rays. He expected radiation to be emitted only after exposure to sunlight, but found that the salts could fog a photographic plate even in darkness. He had discovered a new phenomenon: **radioactivity**.

X-ray of Anna Röntgen's hand
Röntgen's X-ray of his wife's hand shows that the rays penetrated her skin and muscle, but were impeded by denser bones—and her ring.

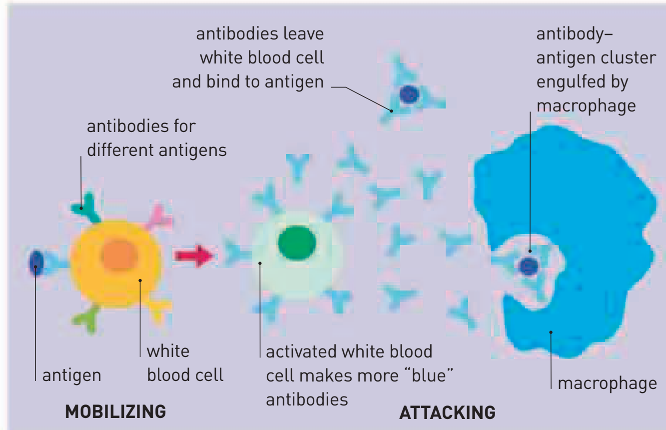
1897

> “ EVERY DAY SEES **HUMANITY** MORE **VICTORIOUS** IN THE STRUGGLE WITH **SPACE AND TIME.** ”
>
> **Guglielmo Marconi, Italian inventor**

IN APRIL, BRITISH PHYSICIST J.J. THOMSON (1856–1940) was studying **cathode rays**. These rays are produced by the negative electrode (cathode) of an electrically charged vacuum tube, and are attracted to the positive electrode (anode). They cause the glass at the far end of the tube to glow. Thomson demonstrated that the rays were composed of particles much lighter than the smallest atoms. He concluded that these "corpuscles," as he called them, were negatively charged components present in all atoms; Thomson had discovered the **first subatomic particles**. They were later called **electrons**.

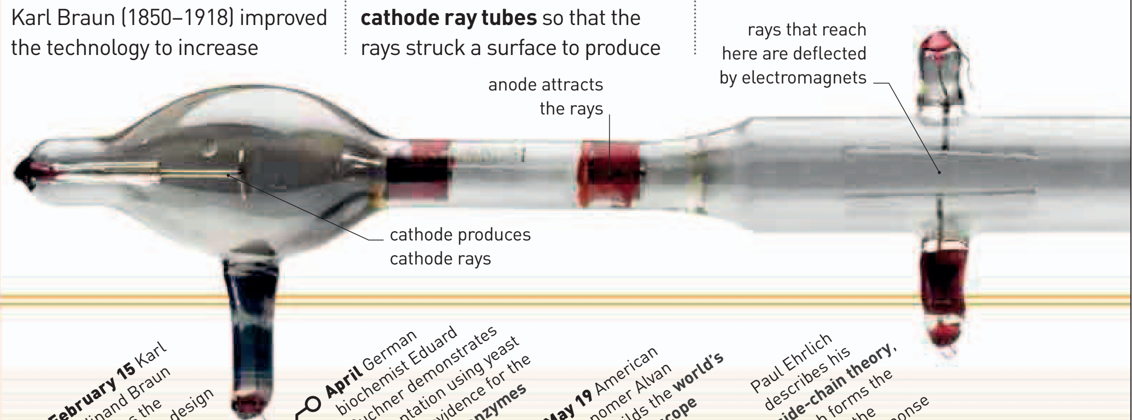
In May of the same year, the first radio communication over water was made across Britain's Bristol Channel. Italian inventor **Guglielmo Marconi** (1874–1937) had been experimenting with **wireless technology**, and in 1897, his team of scientists succeeded in sending a **Morse code signal** from Flat Holm Island to a receiver on the Welsh coast. Later, German physicist Karl Braun (1850–1918) improved the technology to increase broadcasting range. Marconi and Braun went on to share the Nobel Prize 12 years later for their work on wireless radio.

In 1897, Braun was also working on vacuum tubes. He **modified cathode ray tubes** so that the rays struck a surface to produce images. Braun's tube was the first **oscilloscope**—a device that made graphical presentations of electrical signals. It paved the way for the invention of the television and the development



ANTIGEN–ANTIBODY INTERACTION

Paul Ehrlich explained how the immune system could be mobilized to destroy infection. White blood cells carry side-chain antibodies, which bind themselves to foreign particles called antigens. As they are bound together, the white cells are prompted to produce more antibodies. These then cluster around the antigens and enable macrophages—other immune system cells—to destroy them.



1895 Eugenius Warming publishes *Plantesamfund*, classifying **vegetation formations**

November 8, 1895 Wilhelm Röntgen discovers **X-rays**

January 12, 1896 American physician Emil Grubbe describes treating **breast cancer with X-rays**

March 1, 1896 Henri Becquerel accidentally discovers **radioactivity**

March 1896 J.J. Thomson describes how **X-rays** can **ionize** (charge up) gas

May 1896 Ronald Ross correctly suggests that **malaria** is transmitted by **mosquito bites**

February 15 Karl Ferdinand Braun describes the **oscilloscope** design

April German biochemist Eduard Buchner demonstrates fermentation using yeast extract—evidence for the **existence of enzymes**

April 30 J.J. Thomson explains that **cathode rays** are composed of negative particles

May 13 Guglielmo Marconi sends the **first wireless radio signal over water**

May 19 American astronomer Alvan Clark builds the **world's biggest telescope**

Paul Ehrlich describes his **side-chain theory**, which forms the basis of the immune response

August 10 German chemist Felix Hoffmann makes acetylsalicylic acid, later marketed as **aspirin**